PRANAV SARAVANAN

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EDUCATION

Bachelor of Science, Data Science May 2023
Fowler School of Engineering, Chapman University, Orange, CA
Languages: Python, SQL, Scala, Java, JavaScript, C++, GraphQL
ML & Data: TensorFlow, PyTorch, XGBoost, Scikit-learn, Spark ML, PySpark, Spark, Hive
Cloud & Infra: AWS (SageMaker, Athena, Glue, S3, DynamoDB), Databricks (Unity Catalog, MLOps Stacks), Terraform, Kubernetes, Docker
ML Platforms: MLflow, Feast, Airflow (MWAA), Liquibase, Spinnaker, Jenkins, CI/CD
Web & APIs: FastAPI, React, Node.js, Apache Kafka, Redis, LangChain, FAISS
Observability: Prometheus, Grafana, Langfuse, RAGAS, InfluxDB, Git

PROFESSIONAL EXPERIENCE

Disney Entertainment & ESPN Technology, San Francisco, CA Dec 2024 – Present
Machine Learning Engineer, Ad Platforms

- Architecting end-to-end self-service ML platform (Feast, Unity Catalog, DynamoDB, MLflow, Databricks) covering feature stores, schema migrations, online inference, and automated champion deployment across Hulu and Disney+ ad products.
- Built standardized inference container with canary rollout (1%→100% traffic splits), SLO-based auto-rollback, and Airflow-orchestrated daily scoring and retraining pipelines serving production ad-targeting models.
- Developed scalable monitoring agent framework and data-driven dashboards tracking model drift, prediction latency, and SLO compliance across deployed ML services.
- Recovered ~1.8M missing Hulu-on-Disney+ users by diagnosing upstream identity pipeline gaps and coordinating cross-team production fix.
- Reduced 30-day XGBoost training runtime by ~40% via GPU acceleration (Databricks CUDA) with no performance degradation.
- Consolidated three legacy feature store repos into one unified framework, cutting maintenance overhead by 66%.

Chapman University, Orange, CA May 2023 – Dec 2023
ML & Assistive Technology Lab Research Assistant

- Engineered Python-based VR research environments supporting 200+ users; optimized sensor-to-database pipelines for downstream ML training.
- Led cross-platform mobile app development with real-time feedback loops, improving accessibility across research cohorts.

INTERNSHIPS

Pillar, Mountain View, CA May 2022 – Dec 2022
ML Engineer Intern

- Built and deployed a regression model (Scikit-Learn, Node.js) for renovation cost estimation, improving accuracy by 18% with sub-200ms API latency.
- Prototyped a computer vision pipeline (PyTorch) to auto-categorize job-site photos, reducing manual tagging time by 40%.
- Analyzed user behavior (SQL, Python) to identify UX friction points, driving a 15% increase in engagement metrics.

Seedstages, Claremont, CA Mar 2021 – Dec 2021
ML Engineer Intern

- Designed the company's first automated ML pipeline with repeatable workflows for training, validation, and deployment.
- Built real-time feature store for contractor-client matching, improving API response times by 15%.
- Developed NLP text-classification service achieving 15% improvement in automated tagging accuracy (Scikit-learn, NLTK).

PROJECTS

RAG Ops — Football Intelligence Platform (Python, LangChain, FAISS, FastAPI, Langfuse, RAGAS, Docker)

- Production RAG pipeline with Langfuse observability, RAGAS evaluation (faithfulness 0.874), cost tracking ($0.0012/query), FIFA-aesthetic UI — 34 TDD tests, deployed on Railway.

Stream Vault — Streaming Feature Store (Python, Feast, Kafka, Redis, Delta Lake, Docker)

- Dual-write consistency (Delta Lake offline, Redis online), automated parity validation, CI-integrated drift detection, sub-100ms online retrieval.

Sentinel — ML Serving Pipeline (Python, MLflow, FastAPI, Docker, Kubernetes, Prometheus, Grafana)

- Containerized ML serving with MLflow registry, canary rollout (1%→100%), SLO-based auto-rollback, real-time drift monitoring via Evidently AI.

ABOUT ME

- Languages: English (fluent), Tamil (fluent), Spanish (intermediate)